1.10. No. An example against:

$$\Gamma_2 = \{\alpha\}, \quad \Gamma_1 = \{\alpha, \neg\alpha\}$$
$$\Gamma_1 \cap \Gamma_2 = \{\alpha\}$$

1.2. Assume $\Gamma_1 \cup \Gamma_2$ is consistent.
Then there exists a valuation v s.t ~~the~~ every formula under $\Gamma_1 \cup \Gamma_2$ is true.
Because every formula in Γ_2 is in $\Gamma_1 \cup \Gamma_2$, Γ_2 is also consistent under v.
So, $\Gamma_1 \cup \Gamma_2$ is only consistent if Γ_2 is consistant.

3.10.

1.	$\exists y \forall x \forall z R(x,y,z)$	Given	
2.	$\forall x \forall z R(x,b,z)$	R6	$y=b$
3.	$\forall z R(a,b,z)$	R5	$x=a$
4.	$\exists y \forall z R(a,y,z)$	R7	$b=y$
5.	$\exists x \exists y \forall z R(x,y,z)$	R7	$a=x$

?

$R(x,y,z) \leftrightarrow Even(x)$.
Let $x=2$, then $R(x,y,z) \to T$.
α is false because $\forall x, \neg R(x,y,z)$ isn't true.
Therefore β doesn't imply α.

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4.x. i) $\alpha \to (\beta \to \alpha) \equiv T$

1. $\alpha \vee \neg\alpha$ A
2. $\neg\alpha \vee \alpha$ E10
~~3. $\alpha \to \alpha$ E20~~
3. $\neg\alpha \vee (\alpha \vee \neg\beta)$ I1
4. $\neg\alpha \vee (\neg\beta \vee \alpha)$ E10
5. $\neg\alpha \vee (\beta \to \alpha)$ E20
6. $\alpha \to (\beta \to \alpha)$ E20

■

ii) ~~$\alpha \vee \neg\alpha$~~ A $(\alpha \to (\beta \to \gamma)) \to ((\alpha \to \beta) \to (\alpha \to \gamma))$

1. $\alpha \vee \neg\alpha$ A
2. Assume $\alpha \to (\beta \to \gamma)$ R3
3. Assume α R3
4. $\beta \to \gamma$ MP 2,3
5. $\neg\beta \vee \gamma$ E20
6. $(\neg\beta \vee \alpha) \vee (\neg\alpha \vee \gamma)$ I1
7. $\neg(\neg\beta \vee \alpha) \to (\neg\alpha \vee \gamma)$ E20
8. $(\beta \vee \neg\alpha) \to (\neg\alpha \vee \gamma)$ E9
9. $(\neg\alpha \vee \beta) \to (\neg\alpha \vee \gamma)$ E10
10. $(\alpha \to \beta) \to (\alpha \to \gamma)$ E20

iii) $\vdash (\neg\beta \to \neg\alpha) \to ((\neg\beta \to \alpha) \to \beta)$

1. $(\neg\beta \to \neg\alpha)$ I11
2. $(\neg\beta \to \alpha)$ I11
3. $(\beta \vee \neg\alpha)$
4. $(\beta \vee \alpha)$
5. $(\beta \vee \neg\alpha) \wedge (\beta \vee \alpha)$
6. $\beta \vee (\neg\alpha \wedge \alpha)$ $((\beta \vee F) = \beta)$
~~7. $\neg\alpha \wedge \alpha$~~
7. β

β 1-3

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$K_n \Rightarrow (2) = |E|$

4.2. Prove $(\neg\alpha \to \neg\beta) \to (\beta \to \alpha)$

1. Assume $(\neg\alpha \to \neg\beta)$ to prove $(\beta \to \alpha)$
2. Assume (β) to prove (α) IR3
3. $(\neg\alpha \to \neg\beta) \to ((\neg\alpha \to \beta) \to \alpha)$ A3 w/ $\alpha := \alpha, \beta := \beta$
4. $((\neg\alpha \to \beta) \to \alpha)$ MP w/ 1,3
5. $(\beta \to (\neg\alpha \to \beta))$ A1 w/ $\alpha := \beta, \beta := \neg\alpha$
6. $(\neg\alpha \to \beta)$ MP w/ 2,5
7. α MP w/ 4,6
8. $(\beta \to \alpha)$

5.1. 1C. $\forall x\,[x > 1 \to x^2 > x]$

2C. $\forall x\,[x > x^2 \to \forall y\,(x \geq y)]$

3C. $\exists x\,[x < x^2]$

5.2. Let $G(x,y)$ mean "$x > y$", let $E(x,y)$ mean "$x = y$", let $sq(x) = x^2$

1C. is True. If $x > 1$ then $x(x-1) > 0 \Rightarrow x^2 - x > 0 \Rightarrow x^2 > x$

2C. is false. Take $x = \frac{1}{2}$, $\frac{1}{2} > \frac{1}{4}$ but if $y = 1$ the $x < y$.

3C is True. Take $x = 2$, $2 < 4$.

6.1. $\overset{\text{vertices}}{n} + \overset{\text{faces}}{f} - \overset{\text{edges}}{m} = 2$

Let G be bipartite with parts A, B. Any cycle alternates $A, B, A, B, \ldots$ so it's length must be even. True.

6.2. Let G be a forest. Then by definition, G has no cycles. Bipartite is a graph with no odd cycles. Therefore if a graph G is a forest, its also bipartite. True.

6.2. Take K_3, all degrees are 2 but it is not bipartite.

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